

VEILED-Z Gate — Conditioned-Response Suspension Near Spectral Threshold

Status: HYPOTHESIZED structure, built from standard non-Hermitian physics results, not freshly derived in full here. Flagged where calibration is still open.

1. Why this gate exists (the derivation, not just the caution)

The Petermann factor $K = \frac{\langle \psi_L | \psi_L \rangle \langle \psi_R | \psi_R \rangle}{|\langle \psi_L | \psi_R \rangle|^2}$ diverges at an exceptional point (Quantum Criticality paper, eq. 460–461). In non-Hermitian/laser physics this divergence has a known scaling: for a second-order EP, as the system sits a distance ϵ away from criticality,

$$\Delta\lambda \sim \sqrt{\epsilon} \qquad \text{(eq. 458, his own result)}$$
$$K \sim \epsilon^{-1/2} \qquad \text{(standard result for second-order EPs in non-Hermitian systems –}$$

not derived fresh here, cited from the established physics, not asserted)

So K and the eigenvalue splitting share the same $\sqrt{\epsilon}$ family, just inverted. Practically: **the same proximity to threshold that makes a spectral reading most diagnostic also makes it least reliable**, by a known, non-arbitrary rate — not a vague "be careful near the edge."

2. The trigger — VEILED-Z(t)

Using $Z(t) := \alpha(t)$ (the corrected stability functional) as the proxy for distance-to-criticality:

$$\text{VEILED-Z}(t) := \text{TRUE} \quad \text{when} \quad |Z(t)| < Z_{\text{threshold}}$$

$Z_{\text{threshold}}$ should be set by choosing an acceptable amplification ceiling K_{crit} (e.g., "don't trust a reading once amplification exceeds 10×") and solving for the corresponding Z . **This calibration step is not done here** — it needs either simulation data (the same data Prong 3 is already waiting on) or a tighter derivation linking $Z(t)$ directly to $K(t)$ for the specific operator in use. Until then, treat $Z_{\text{threshold}}$ as a placeholder to be set, not a number this document supplies.

3. The gate behavior

$$\text{IF VEILED-Z}(t) = \text{TRUE:}$$

SUSPEND any conditioned-response rule that would fire directly off $Z(t)$

HOLD execution open, pending one of:

- (a) an independent confirmation signal not subject to the same EP-driven amplification, or
- (b) time-averaging sufficient to reduce the amplified noise, or
- (c) explicit deliberate review rather than automatic firing

DEFAULT (while held): the One-Good baseline action — the option that does the least harm if the suspended reading turns out to be wrong in either direction

ELSE:

Normal conditioned-response behavior applies

"Held open" means specifically: the system does not act on $Z(t)$ as if it were trustworthy, and does not act as if it had no information either — it falls back to a pre-agreed safe default until the zone clears or independent confirmation arrives.

4. Where this sits in the existing architecture

This isn't a new category — it's a specific, derivable instance of the **Runtime Containment Requirement (RCR)** already in EAAP: a containment trigger whose firing condition is now tied to a real, named physical mechanism (EP-driven eigenvector non-orthogonality) rather than a general caution principle. Framing it that way keeps it inside the existing gate architecture rather than adding a parallel one.

5. Status

Sound, derivation-backed:

- The reason the gate is needed (Petermann divergence) is real and correctly attributed
- The qualitative behavior (suspend, hold, default to least-harm) is a standard, well-justified circuit-breaker pattern

Open:

- $Z_{\text{threshold}}$ is uncalibrated — needs either data or a tighter derivation
- The exact $K(\epsilon)$ scaling exponent for *this* operator (cardiac $L(t)$, not a generic 2-level laser model) hasn't been independently confirmed — borrowed from the general non-Hermitian-physics result, not re-derived for this specific case